Math Bootcamp: Content to be Covered

For Incoming MSc Economics Students

Math Camp 2026

Overview

- **Objective:** Bridge the gap between a weaker undergraduate math background and MSc-level economics courses.
- **Duration:** 3 hours per day for 5 days.
- **Method:** Focus on intuition, fundamentals, and confidence-building. Rigorous proofs and advanced techniques are deferred to fall courses; only simple proofs are included where necessary.
- **Emphasis:** Conceptual topics to be covered throughout graduate microeconomics, macroeconomics, econometrics, and mathematical economics.

Contents

1 Day 1 — Foundations (Sections 1.1–1.9)

Approximate duration: 3 hours. See Day 1 slides (Sections 1.1–1.9).

1.1 1.1 — Notation and Number Systems

- Number systems: $\mathbb{N}$ (counting), $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{R}^n$; nesting $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$.
- Symbols: $\in$, $\subset$, $\forall$, $\exists$, $\Rightarrow$, $\Leftrightarrow$.
- Scalars versus vectors: $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$; dot product $\mathbf{p} \cdot \mathbf{x} = \sum_i p_i x_i$.
- Practice: identify the smallest set containing a given object (e.g. counts in $\mathbb{N}$; prices in $\mathbb{Q}$; order vectors in product sets).

1.2 1.2 — Logic and Mathematical Statements

- P **only if** Q ($P \Rightarrow Q$): Q necessary for P.
- P **if** Q ($Q \Rightarrow P$): Q sufficient for P.
- P **iff** Q ($P \Leftrightarrow Q$): necessary and sufficient.
- Contrapositive: $\neg Q \Rightarrow \neg P$ equivalent to $P \Rightarrow Q$.
- Running example: IKEA budget set $\mathcal{B} = \{\mathbf{x} \in \mathbb{N}^3 : \mathbf{p} \cdot \mathbf{x} \leq B\}$; practice with $\in$, $\subset$, $\forall$, $\exists$, $\Rightarrow$, and contrapositive.

1.3 1.3 — Sets and Set Operations

- Notation: $\in$, $\notin$, $\subset$, $=$, $\emptyset$, roster and set-builder forms.
- Union, intersection, complement, difference; Cartesian product $A \times B$.
- Economic sets: budget sets, consumption sets; product spaces such as $\mathbb{Q} \times \mathbb{N}$ versus constraint sets like $\mathcal{B}$.
- Practice: describe $\mathcal{B} \cap \mathcal{H}$, $\mathcal{B} \cup \mathcal{H}$, $\mathcal{B} \setminus \mathcal{H}$; distinguish elements from subsets.

1.4 1.4 — Functions

- Function $f \colon A \to B$: domain, codomain, range $f(A) \subseteq B$.
- Injective (one-to-one) and surjective (onto); codomain versus range.
- Operations: $(f \pm g)$, (cf), (fg), composition $(f \circ g)(x) = f(g(x))$.
- A rule must assign **exactly one** output per input (not a correspondence).
- Practice: domain, codomain, range, injectivity, and surjectivity (e.g. partner age as a function of your age).

1.5 1.5 — Supremum and Infimum

- Example first: reservation price / willingness to pay when the purchase set $P = \{p \in \mathbb{R}_+ : p < \bar{p}\}$ has no maximum but $\sup P = \bar{p}$.
- $\max S$ / $\min S$ versus $\sup S$ / $\inf S$ (least upper bound, greatest lower bound).
- When extrema are attained versus only approached (e.g. $S = [0, 1)$, $\sup S = 1 \notin S$).
- If $\max S$ exists, $\max S = \sup S$; always $\inf S \leq x \leq \sup S$ for $x \in S$.

1.6 1.6 — Algebra Revision

- Taylor expansion around a: $f(a+h) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} h^k$ (first-order: $f(a+h) \approx f(a) + f'(a)h$).
- Log and exponential rules; series $\ln(1+x)$ and e^x.
- **Why logs:** $\Delta \ln x \approx \Delta x/x$ tracks percentage changes; elasticity $\varepsilon_{y,x} = \partial \ln y / \partial \ln x$; log–log regression.
- Cobb–Douglas: $U = x_1^{\alpha} x_2^{1-\alpha}$, $Y = AK^{\alpha} L^{1-\alpha}$; linear in logs; constant elasticities and expenditure shares.
- Local log approximation of any smooth $u > 0$ near $\bar{\mathbf{x}}$; Cobb–Douglas as a modeling choice when used globally.
- Homogeneity of degree k: $f(t\mathbf{x}) = t^k f(\mathbf{x})$; Cobb–Douglas utility and production are degree 1 (CRS: double inputs $\Rightarrow$ double output).

1.7 1.7 — Single-Variable Derivatives

- Formal definition: $f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$; meaning as slope and marginal rate (MC, MU, MP_K).
- Rules: constants, sums, power, product, quotient, chain rule; $(e^x)' = e^x$, $(\ln x)' = 1/x$, $(a^x)' = a^x \ln a$; inverse-function rule (stated).
- Practice: cubic cost, monopolist profit, Cobb–Douglas $Y(K)$, composite cost (chain rule).
- **Euler's theorem** (for C^1 homogeneous f of degree k): $\sum_i x_i f_{x_i} = kf(\mathbf{x})$ (proof via $t \mapsto f(t\mathbf{x})$).
- Adding-up problem: under CRS and competitive factor markets, $PQ = rK + wL$; IRS ($k > 1$) and DRS ($k < 1$) and payment exhaustion.

1.8 1.8 — Multivariate Calculus

- Level curves $\mathcal{L}_c = \{(x, y) : f(x, y) = c\}$ and contour maps; 3D slice picture for $z = f(x, y)$.
- Roadmap: partials $\Rightarrow$ total differential $\Rightarrow$ directional derivative $\Rightarrow$ cross-partials (Schwarz).
- Partial derivatives (limit definition; fix one variable); $MP_K = \partial Y / \partial K$ with L fixed.
- Total differential: add/subtract trick, MVT on each bracket, $dz = f_x\, dx + f_y\, dy$; Cobb–Douglas output example $dY = MP_K\, dK + MP_L\, dL$.
- Gradient ∇f: perpendicular to level curves; points uphill; $D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u} = \|\nabla f\| \cos\theta$.
- Directional derivative along unit direction $\mathbf{u}$; example: subsidy requiring input changes in ratio 2:1.
- Cross-partials and Schwarz's theorem (stated); Y_{KL} and complements versus substitutes.

1.9 1.9 — Limits

- Motivation: Berkeley's critique of "infinitely small" h in early calculus.
- Sequence limit (ε–N): quantifier order and convergence.
- Function limit (ε–δ): $0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon$; example $f(x) = x + 1$ for $x \neq 1$, $f(1) = 3$, $\lim_{x \to 1} f(x) = 2 \neq f(1)$.

- Continuity at a: $\lim_{x\to a} f(x) = f(a)$; differentiability at a: derivative limit exists.
- Differentiable $\Rightarrow$ continuous; converse fails ($|x|$ at 0; Weierstrass function).
- Practice: ε–δ proofs for $f'(a) = 2a$ when $f(x) = x^2$ and for $\lim_{x\to 1} f(x) = 2$.

1.10 Appendix (Slides)

- Derivations of single-variable rules: constants, sums, powers, product, quotient.
- Key limits for $(e^x)'$ and $(\ln x)'$; exponential and logarithm derivative proofs.

2 Day 2 — More Calculus and Real Analysis Bridge

Approximate duration: 3 hours. See Day 2 slides (Sections 2.1–2.6).

2.1 2.1 — Implicit Function Theorem

- Explicit versus implicit relationships; when $F(x, y) = 0$ cannot be solved in closed form.
- Comparative statics: $F(x, \theta) = 0$ defines $y^*(\theta)$; goal is $dy^*d\theta$.
- **Core recipe:** totally differentiate $F(x, y) = 0$ to get $dydx = -F_xF_y$ when $F_y \neq 0$.
- Regularity: $F_y \neq 0$ means finite slope locally; vertical tangent when $F_y = 0$.
- **Practice (Together):** nonlinear demand $D(p, \alpha) = \alpha e^{-p}$, supply $S(p) = p$; no closed form for $p^*(\alpha)$, but IFT still gives $dp^*d\alpha$.
- **Applications:** market clearing $E(p, \alpha) = D(p, \alpha) - S(p) = 0$; linear example checks IFT against explicit $p^*(\alpha) = \alpha - cb + d$.
- Direct versus indirect effects; chain-rule derivation of $dp^*d\alpha = -E_\alpha E_p$.
- **Tax incidence:** per-unit tax on sellers; $dp^*dt = db + d$; tax borne more heavily by the **less elastic** side.
- **Appendix (for interest):** several-variable / Jacobian form of IFT.

2.2 2.2 — EVT, MVT, and Existence

- **Open versus closed sets** (budget, strict budget, reservation price examples).
- EVT on closed interval $[a, b]$: continuous f attains max and min.
- When EVT fails: not closed, not bounded, or not continuous (failure figure).
- **Compact** in $\mathbb{R}^n$ (camp level): closed and bounded (Heine–Borel, stated).
- **Weierstrass EVT:** continuous f on compact K attains max and min.
- **Bridge:** compact feasible set + continuous objective $\Rightarrow$ optimum exists.
- Rolle's theorem; MVT; labor/output economics example.
- **Appendix (for interest):** Weierstrass EVT proof idea.

2.3 2.3 — Integration

- Riemann integral as signed area / accumulation; definite versus indefinite integrals.
- **For interest:** Lebesgue integral (measures level sets; handles sparse jumps).
- Fundamental Theorem of Calculus: $F'(x) = f(x) \Rightarrow \int_a^b f(x)\,dx = F(b) - F(a)$.
- Integration by parts: $\int u\,dv = uv - \int v\,du$ (product-rule proof); example $\int xe^x\,dx$.
- **Probability link:** PMF and CMF (discrete dice); PDF and CDF (continuous Unif$(0,1)$); FTC link $F'(x) = f(x)$.
- Expectation for non-negative continuous X: $\mathbb{E}[X] = \int_0^\infty xf(x)\,dx = \int_0^\infty (1 - F(x))\,dx$.
- **Appendix (for interest):** expectation-from-CDF proof (integration by parts).

2.4 2.4 — Leibniz Rule and Fubini

- **Fixed limits:** if $F(\theta) = \int_a^b f(x,\theta)\,dx$, then under regularity $dFd\theta = \int_a^b \partial f\partial\theta(x,\theta)\,dx$.
- Regularity (camp level): $\partial f/\partial\theta$ exists and f, $\partial f/\partial\theta$ are continuous on $[a,b] \times I$.
- **Variable limits:** when $a(\theta)$, $b(\theta)$ move, $F'(\theta)$ adds endpoint terms plus the integral of $\partial f/\partial\theta$.
- When fixed-limit Leibniz fails: moving discontinuity (step function) example.
- **Fubini:** if $_D|f(x,y)|\,dx\,dy < \infty$, then $_Df = \int\int f$ (order of integration can be swapped).
- Camp shortcut: bounded continuous f on a nice region; Tonelli (for interest) when $f \geq 0$.
- Example: $_D(x+y)\,dx\,dy$ on triangle $D = \{0 \leq x \leq 1,\ 0 \leq y \leq x\}$.

2.5 2.5 — Taylor Expansion and Linearization

- One-variable Taylor through order n; multivariable C^2 form with ∇f and Hessian H_f.
- First-order truncation: $f(\mathbf{a}+\mathbf{h}) \approx f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot \mathbf{h}$.
- Second-order truncation: adds $12\,\mathbf{h}^\mathsf{T} H_f(\mathbf{a})\,\mathbf{h}$.
- Two-variable explicit formula from matrix multiplication (optional if Hessian is familiar).
- When linearization works: small shifts, modest curvature, differentiability; fails with large shifts, kinks, strong nonlinearity.
- **Practice (Together):** first-order two-variable Taylor for $f(x,y) = x^2 + y$ at $(1,1)$.
- **Take home:** second-order two-variable Taylor (e.g. $f(x,y) = \sin x + \cos y$ at $(0,0)$).
- **Appendix (for interest):** one-variable Taylor proof; Itô's lemma and Brownian motion.

2.6 2.6 — Uniform Convergence

- Pointwise convergence: $f_n(x) \to f(x)$ for each fixed x.
- Uniform convergence: one N works for all x in the domain.

- Example: $f_n(x) = x^n$ on $[0, 1]$ converges pointwise but not uniformly.
- **Why it matters:** uniform convergence justifies swapping limits with continuity, integration, and (with extra conditions) differentiation.
- Link to section 2.4: Leibniz/Fubini need regularity so limits and integrals can be interchanged safely.

2.7 Appendix (Day 2 Slides)

- IFT — several variables (Jacobian form).
- Weierstrass EVT — proof idea.
- Expectation from the CDF (integration by parts).
- Leibniz rule failure (moving step).
- Taylor expansion proof (one variable); Itô's lemma and Brownian motion (for interest).

3 Day 3 — Linear Algebra and Real Analysis

*Approximate duration: 3 hours. Linear algebra is **must cover**; real-analysis topics below are prioritized.*

3.1 Linear Algebra — must cover (Block 1)

- **Vectors in $\mathbb{R}^n$:** notation $\mathbf{x} = (x_1, \ldots, x_n)$; dot product $\mathbf{a} \cdot \mathbf{b} = \sum_i a_i b_i$ (expenditure $\mathbf{p} \cdot \mathbf{x}$, projections).
- **Matrices:** dimensions $m \times n$, entries, matrix–vector product $A\mathbf{x}$; matrix multiplication (order matters).
- **Transpose** A^T.
- **Linear systems:** $A\mathbf{x} = \mathbf{b}$; solution as intersection of hyperplanes (economic: market clearing, input–output balance).
- **Determinant** (must know 2×2 formula): $\det A \neq 0 \Leftrightarrow A$ invertible (square case).
- **Inverse** A^{-1}: $AA^{-1} = I$; solve $\mathbf{x} = A^{-1}\mathbf{b}$ when it exists.
- **Linear independence** (intuitive): no vector in a set is a linear combination of the others; dependence when one column is redundant.
- **Rank** (conceptual): number of linearly independent rows/columns; full rank $\Leftrightarrow$ invertible for square A.

3.2 Convergence — if time allows (Day 3)

- Deeper practice with uniform convergence (Day 2.6 is the core statement).
- Additional counterexamples and exchange-of-limit applications.

3.3 Real Analysis — if time allows (Day 3)

- **Separation theorem** (picture-based): disjoint convex sets separated by a hyperplane; supporting hyperplanes; welfare economics and Pareto efficiency (high level).

- **Fixed point theorems** (statement): equilibrium as a fixed point; Brouwer or contraction mapping; existence of equilibrium (no proof).
- **Continuity recap:** ε–δ at a point versus on a set; continuity of demand, supply, profit functions; when discontinuities break optimization.
- **Product spaces** $\mathbb{R}^n$: choice sets as subsets of product domains (brief, if not done Day 1).
- **Metric spaces / neighborhoods** (for interest only): distance, open balls; optional rigor beyond open/closed sets from Day 2.

4 Day 4 — Optimization

*Approximate duration: 3 hours. Convex **sets** (feasible regions) are covered on Day 3 — callback only. Focus: FOC/SOC, concave/quasiconcave **functions**, Lagrange, envelope, geometric Farkas $\Rightarrow$ KKT.*

4.1 4.1 — Unconstrained Optimization (1D, then multivariable)

- **Single variable:** local versus global optima; interior FOC $f'(x^*) = 0$.
- **SOC via concavity:** concave $\Rightarrow$ first-order Taylor above f; at $f'(x^*) = 0$, local max. Convex dual for min. Figure: tangent above/below graph.
- $f'' \leq 0 \Leftrightarrow$ **concave** (C^2): Taylor proof for $\Rightarrow$; symmetric second-difference limit for $\Leftarrow$.
- Remark: $f(x) = -x^4$ strictly concave but $f''(0) = 0$; $f'' < 0$ sufficient, not necessary, for strict concavity.
- **Multivariable:** $\nabla f(\mathbf{x}^*) = \mathbf{0}$; SOC stated via concavity / H_f ND; proof in appendix.
- **Together:** monopoly $\pi(q) = pq - q^2$. **Take home:** two-input convex cost $C = q_1^2 + q_2^2$.

4.2 4.2 — Concavity, Quasiconcavity, and Global Optima

- **Convex/concave functions** (not convex sets — see Day 3): secant below graph (concave) / above graph (convex); upper and lower contour sets.
- **Quasiconcave** (maximization): along $\mathbf{z} = \lambda\mathbf{x} + (1-\lambda)\mathbf{y}$, $f(\mathbf{z}) \geq \min\{f(\mathbf{x}), f(\mathbf{y})\}$; equivalently upper contour sets are convex.
- **Prove equivalence** quasiconcavity $\Leftrightarrow$ convex upper contour sets (two-direction proof).
- Concave $\Rightarrow$ quasiconcave; Cobb–Douglas utility is quasiconcave (stated).
- **Global sufficiency:** quasiconcave f on convex feasible set $\mathcal{B}$ (Day 3 callback) $\Rightarrow$ interior FOC/Lagrange solution is a global maximum.
- **Together:** Cobb–Douglas quasiconcavity. **Take home:** $u = x_1 x_2$.

4.3 4.3 — Equality Constraints and Lagrange Multipliers

- Problem: $\max f(x, y)$ subject to $h(x, y) = 0$; constraint defines a **curve** in the plane.
- **3D picture:** surface $z = f(x, y)$; constraint $h(x, y) = 0$ is a curve on the domain; optimum on the curve.
- **Tangent argument:** any local direction $\mathbf{d}$ along the curve satisfies $\nabla h \cdot \mathbf{d} = 0$; if $\nabla f \cdot \mathbf{d} \neq 0$, then f increases along the curve locally $\Rightarrow$ not optimal. Hence $\nabla f \parallel \nabla h$.

- Lagrangian (equality): $\mathcal{L} = f + \mu h$; FOC $\nabla f + \mu \nabla h = 0$, $h = 0$.
- μ **unrestricted** (either sign): equality allows both tangent directions.
- Shadow price interpretation; Cobb–Douglas budget (callback Day 1).
- **Together:** budget max with $\mathcal{L} = u + \mu(w - p \cdot \mathbf{x})$. **Take home:** cost min with production equality.

4.4 4.4 — Envelope Theorem

- **Placement:** after Lagrange setup, before inequalities/KKT.
- Value function $V(\theta) = \max_{\mathbf{x}} f(\mathbf{x}; \theta)$ (subject to constraints).
- **Unconstrained:** $dVd\theta = \partial f \partial \theta\Big|_{\mathbf{x}^*(\theta)}$ — no need to differentiate $\mathbf{x}^*(\theta)$.
- **Constrained:** $dVd\theta = \partial \mathcal{L} \partial \theta\Big|_{(\mathbf{x}^*, \mu^*, \lambda^*)}$.
- Economic examples: profit vs. price shifter; indirect utility vs. income (one each).
- Contrast with chain rule through $\mathbf{x}^*(\theta)$: envelope drops terms that vanish at the optimum.
- **Together:** $\pi^*(p)$ envelope gives $d\pi^*/dp = q^*$. **Take home:** $V(p, w)$, $\partial V/\partial w = \lambda$.

4.5 4.5 — Inequalities, Geometric Farkas, and KKT

- Inequalities $g_j(\mathbf{x}) \leq 0$; binding versus slack; complementary slackness $\lambda_j g_j(\mathbf{x}^*) = 0$.
- **Feasible directions** at a boundary point form a **convex cone**: $\nabla g_j \cdot \mathbf{d} \leq 0$ for active constraints.
- No improving feasible direction: for a max, no $\mathbf{d}$ with $\nabla f \cdot \mathbf{d} > 0$ and $\nabla g_j \cdot \mathbf{d} \leq 0$.
- **Geometric Farkas (picture):** in the (x_1, x_2) plane, active ∇g_j span cone K; at optimum $-\nabla f \in K$. Figure: `farkas_cone_2d.png` (Day 4 slides).
- **KKT** (stated): stationarity, primal/dual feasibility, complementary slackness.
- **Sign mnemonic** — Convention A only on slides: $\mathcal{L} = f + \sum_j \lambda_j g_j + \sum_k \mu_k h_k$ (all plus); see table below.
- Nonnegativity $x_i \geq 0$ and corner solutions (conceptual).
- **Sufficiency (brief):** quasiconcave f + convex $\mathcal{B} \Rightarrow$ KKT point is global; constraint qualification (e.g. LICQ) for necessity (name only).
- **Together:** budget + inequality KKT; min cost with $\geq$ constraint. **Take home:** corner $x_1 = 0$.
- **Appendix (for interest):** algebraic Farkas proof; link to `duality_notes`.

4.6 KKT sign mnemonic (reference)

Convention A (fixed form). Use $\mathcal{L} = f + \sum_j \lambda_j g_j$ with the constraint written exactly as in the problem ($g \leq 0$ or $g \geq 0$):

Problem	**Constraint**	λ
$\max f$	$g \le 0$	$\lambda \ge 0$
$\max f$	$g \ge 0$	$\lambda \le 0$
$\min f$	$g \le 0$	$\lambda \ge 0$
$\min f$	$g \ge 0$	$\lambda \le 0$
Equality $h = 0$: $\mathcal{L} = f + \mu h$; μ **unrestricted**.		

Short rule (Convention A): $\le$ on the constraint $\Rightarrow \lambda \ge 0$; $\ge$ on the constraint $\Rightarrow \lambda \le 0$.

Memory tricks (sign of objective $\times$ sign of constraint inequality):

- $\max f$ $(+)$ with $g \le 0$ $(-)$: "opposite" $\Rightarrow \lambda \ge 0$.
- $\min f$ $(-)$ with $g \le 0$ $(-)$: "$- \times -$" $\Rightarrow \lambda \ge 0$.
- $\max f$ $(+)$ with $g \ge 0$ $(+)$: "same" $\Rightarrow \lambda \le 0$.
- $\min f$ $(-)$ with $g \ge 0$ $(+)$: "opposite" $\Rightarrow \lambda \le 0$.

Convention B (footnote only — not used on Day 4 slides). An alternative is to flip the constraint into $\mathcal{L}$ so $\lambda_j \ge 0$ always; see fall-course notes if needed.

4.7 Appendix (Day 4 Slides)

- Multivariate SOC proof (Hessian ND).
- Algebraic Farkas lemma and KKT derivation.
- Bordered Hessian / SOC for constrained problems (for interest only).
- Take-home solutions (§4.1–4.5).

5 Day 5 — Dynamic Programming

Approximate duration: 3 hours.

5.1 Motivation and Framework

- Sequential decision-making over time.
- Intertemporal trade-offs.
- Finite versus infinite horizon (finite focus for bootcamp).
- Markov structure: current state summarizes relevant history.

5.2 State Variables versus Control Variables

- State: variables that evolve according to law of motion.
- Control: variables chosen each period.
- Transition equation or law of motion.
- Feasibility constraints in dynamic problems.
- Classic problem types: consumption–savings, capital accumulation, resource extraction.

5.3 Bellman Equation

- Value function: maximum attainable payoff from a given state.
- Principle of optimality (Bellman's idea).
- Recursive formulation of dynamic problems.
- Bellman equation for finite-horizon problems.
- Terminal condition and backward solution logic.

5.4 Backward Induction

- Solving finite-horizon problems from the last period backward.
- Relationship between period-by-period optimization and Bellman recursion.
- Policy function: mapping from state to optimal control.

5.5 Value Function Iteration (Conceptual)

- Idea of iterating on the Bellman operator.
- Convergence to fixed point of Bellman equation (intuition only, no algorithm).
- Distinction between solving finite-horizon and infinite-horizon problems.

5.6 Two-Period Models as Building Blocks

- Structure of two-period consumption–savings problems.
- Euler equation (conceptual introduction if appropriate).
- Link between dynamic programming and standard intertemporal optimization.

6 Cross-Cutting Themes (All Days)

- Mathematical notation used in economics graduate courses.
- Reading and interpreting first-order conditions, second-order conditions, and comparative statics.
- Connecting mathematical objects to economic meaning (marginal, shadow price, value function).
- Confidence with standard problem templates used in fall courses.

7 Suggested Time Allocation

Day	Block 1 (∼1 hr)	Block 2 (∼1 hr)	Block 3 (∼1 hr)
1	1.1–1.4: notation; logic; sets; functions	1.5–1.6: sup/inf; algebra; Cobb–Douglas	1.7–1.9: derivatives; Euler; multivariate; limits
2	2.1 IFT	2.2 EVT/MVT; 2.3 integration	2.4–2.5; 2.6 uniform conv.
3	Linear algebra (must)	Convergence practice (if time)	Separation; fixed points (if time)
4	4.1 FOC/SOC; 4.2 quasiconcavity	4.3 Lagrange; 4.4 envelope	4.5 Farkas geometry; KKT
5	Framework; state and control	Bellman equation	Backward induction; VFI idea